

Project Design Phase-II

Data Flow Diagram & User Stories

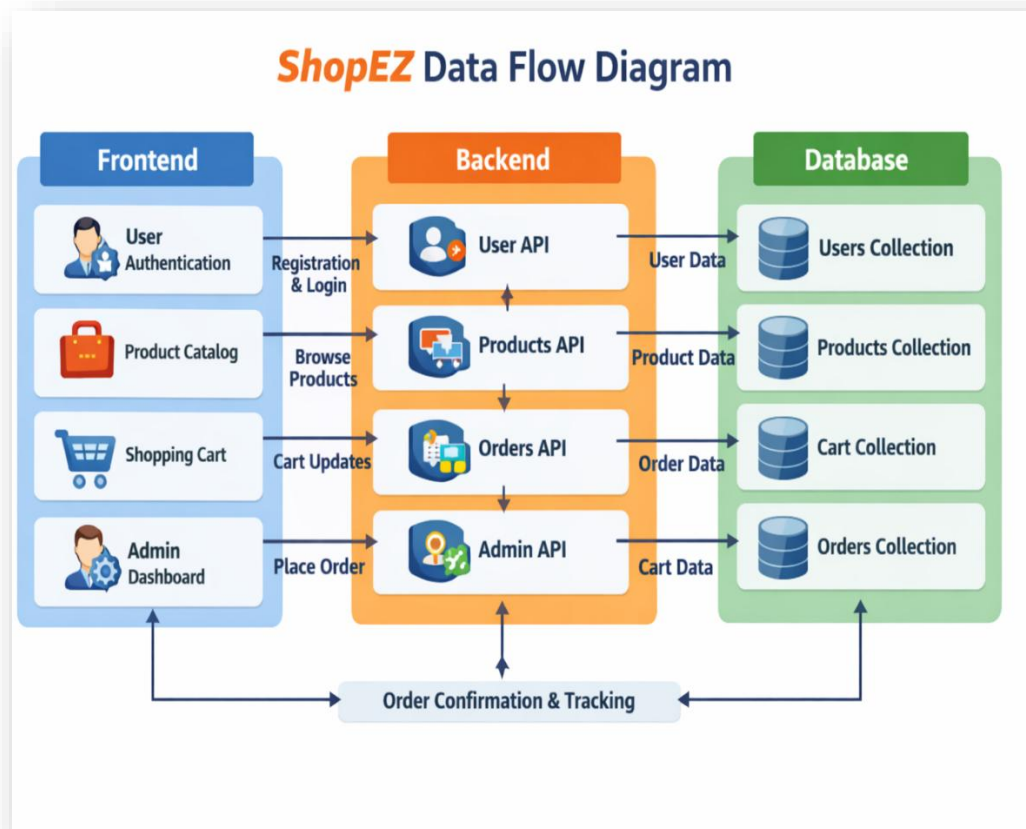
Date	18th June 2026
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Project Name	ShopEZ – E-Commerce Application Using Mern Stack
Maximum Marks	4 Marks

Data Flow Diagrams:

At a high level, Shop-EZ's DFD would look like this:

- **User Registration** → Data flows from customer input (email, password, social login) into the **User Authentication API** → stored in **Users collection** in the database.
- **Product Browsing** → Customer requests product data → Backend fetches from **Products collection** → displayed on frontend catalog with filters, reviews, and discounts.
- **Cart Management** → User adds items → Cart API updates **Cart collection** (linked by User ID).
- **Order Placement** → Checkout form (address, payment) → Order API → stored in **Orders collection** → triggers **Order Confirmation** notification.
- **Admin Dashboard** → Admin inputs banner/categories → Backend updates **Products/Orders collections** → reflected in frontend for customers.

Flow Diagram:



This flowchart clearly shows how **Frontend**, **Backend**, and **Database** interact:

- **User Authentication** → Registration/Login data flows to **User API** → stored in **Users collection**.
- **Product Catalog** → Browsing requests go to **Products API** → retrieved from **Products collection**.
- **Cart & Checkout** → Cart updates and orders flow through **Orders API** → stored in **Cart/Orders collections**.
- **Admin Dashboard** → Admin inputs (banners, categories) update backend and reflect in frontend.
- **Order Confirmation** → Backend sends confirmation & tracking back to frontend.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Mobile/Web)	Registration	USN-1	As a user, I can register with email & password.	Account/dashboard accessible	High	Sprint-1
		USN-2	As a user, I receive confirmation email after registration.	Confirmation email received & verified	High	Sprint-1
		USN-3	As a user, I can register via Facebook.	Facebook login grants dashboard access	Low	Sprint-2
		USN-4	As a user, I can register via Gmail.	Gmail login grants dashboard access	Medium	Sprint-1
	Login	USN-5	As a user, I can log in with email & password.	Successful login redirects to dashboard	High	Sprint-1
	Product Catalog	USN-6	As a user, I can browse products with filters & reviews.	Products displayed with accurate details	High	Sprint-1
	Cart	USN-7	As a user, I can add/remove products in my cart.	Cart updates correctly per user ID	High	Sprint-1
	Checkout	USN-8	As a user, I can securely enter address & payment details.	Order placed successfully	High	Sprint-2
	Order Confirmation	USN-9	As a user, I receive instant order confirmation.	Notification & order details page visible	High	Sprint-2
Seller/Admin	Dashboard	USN-10	As an admin, I can manage products, categories, and banners.	Updates reflected in catalog	High	Sprint-2
	Analytics	USN-11	As a seller, I can view order analytics.	Dashboard shows real-time insights	Medium	Sprint-3

